

Heat, Kinetic Theory and Thermodynamics

Temperature, heat and the laws that govern engines, refrigerators, living bodies and the direction of time.

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Mechanics deals with the ordered motion of whole bodies. **Thermal physics** deals with the disordered motion of their atoms and molecules, and with the energy hidden in that motion. This chapter covers temperature and thermal expansion, heat capacity and phase changes, how heat moves (conduction, convection and radiation), the kinetic theory that links temperature to molecular speeds, and the laws of thermodynamics. These laws limit every engine, power station, refrigerator and living cell. Engineers use them to design heat exchangers and engines, and physiologists use them to understand how the body holds its temperature at 37 °C.

YOU WILL BE ABLE TO

- Use temperature scales, the zeroth law, and linear and volume thermal expansion, including thermal stress.
- Solve calorimetry problems with specific and latent heats.
- Calculate heat flow by conduction (including through layered walls), convection and radiation.
- Use the ideal gas law, and derive pressure and temperature from kinetic theory.
- Apply the first law of thermodynamics to isobaric, isochoric, isothermal and adiabatic processes on pV diagrams.
- Analyse heat engines, refrigerators and heat pumps, the Carnot limit, and entropy.

THINK FIRST

On a cold morning, a metal bench feels much colder than a wooden one, although both have been outside all night and are at the same temperature. Why?

Your skin does not sense temperature directly. It senses how fast **heat flows** out of it. Metal is an excellent thermal conductor (about 1000 times better than wood), so it pulls heat away from your skin quickly, and the skin surface cools fast. Wood conducts poorly, so the skin in contact with it stays warm. Both benches are at the same temperature, but the rates of heat flow differ hugely. This is conduction, $P = kA \Delta T / L$, in action.

§12.1

Temperature and thermal equilibrium

Two bodies in thermal contact exchange energy until they reach **thermal equilibrium**. Temperature is the property that is then the same for both.

KEY IDEA

Zeroth law of thermodynamics. If bodies A and B are each in thermal equilibrium with a third body C (a thermometer), then A and B are in thermal equilibrium with each other. This is what makes temperature measurable.

Temperature scales. The SI scale is the **kelvin** (K). It is an absolute scale, with zero at **absolute zero**, where the thermal motion of molecules is at its minimum. The Celsius scale uses the same size of degree:

$$T(\text{K}) = T(^{\circ}\text{C}) + 273.15, \quad T(^{\circ}\text{F}) = \frac{9}{5}T(^{\circ}\text{C}) + 32.$$

A temperature *difference* has the same size in K and in $^{\circ}\text{C}$. In gas laws, radiation, efficiencies and entropy, always use kelvin.

§12.2

Thermal expansion

Most materials expand when heated, because their atoms vibrate with larger amplitude in an asymmetric potential well, so their average separation grows. For a length L_0 :

$$\Delta L = \alpha L_0 \Delta T.$$

For a volume V_0 :

$$\Delta V = \beta V_0 \Delta T, \quad \beta \approx 3\alpha \text{ for solids.}$$

Typical values: steel $\alpha = 12 \times 10^{-6} \text{ K}^{-1}$, aluminium 23×10^{-6} , ordinary glass 9×10^{-6} , Pyrex 3×10^{-6} , tooth enamel about 11×10^{-6} .

Thermal stress. If a rod is prevented from expanding, it develops a stress equal to the stress needed to compress it back by $\alpha \Delta T$:

$$\sigma = Y\alpha \Delta T.$$

Water is anomalous. It contracts on heating from 0°C to 4°C , so it is densest at 4°C . Lakes therefore freeze from the top down, and the dense 4°C water at the bottom lets aquatic life survive the winter.

WORKED EXAMPLE 12.1

Bridges and rails

(a) A steel bridge deck is 500 m long. By how much does its length change between -10°C in winter and 40°C in summer? (b) A steel rail is clamped at both ends at 10°C , so that it cannot expand. Find the stress in it at 40°C . ($Y = 200 \text{ GPa}$.)

Solution

(a) $\Delta L = \alpha L_0 \Delta T = (12 \times 10^{-6})(500)(50) = 0.30 \text{ m}$. Expansion joints totalling 30 cm must be built into the deck.

(b) $\sigma = Y\alpha \Delta T = (2.0 \times 10^{11})(12 \times 10^{-6})(30) = 72 \text{ MPa}$. That is a large compressive stress. In a long continuous rail it can make the track buckle sideways into “sun kinks” (Problem P12.17). Modern continuous welded rail is therefore laid at a carefully chosen “neutral” temperature and restrained by heavy sleepers.

§12.3

Heat, heat capacity and phase changes

Heat Q is energy transferred because of a temperature difference. It is measured in joules. The older unit is the calorie: $1 \text{ cal} = 4.186 \text{ J}$, and the food “Calorie” is 1 kcal. Heat is energy *in transit*. A body does not “contain” heat; it contains internal energy.

To raise the temperature of a mass m by ΔT requires

$$Q = mc\Delta T,$$

where c is the **specific heat capacity**, in J/(kg K). Water has an unusually large value, $c = 4186 \text{ J/(kg K)}$. Other values: ice 2100, aluminium 900, iron 450, copper 385, the human body about 3500.

At a **phase change** (melting, boiling) the temperature stays constant while energy goes into breaking intermolecular bonds:

$$Q = mL,$$

where L is the **latent heat**. For water, $L_f = 3.34 \times 10^5 \text{ J/kg}$ (fusion, at 0°C) and $L_v = 2.26 \times 10^6 \text{ J/kg}$ (vaporisation, at 100°C). Near body temperature the latent heat of vaporisation is about $2.4 \times 10^6 \text{ J/kg}$.

Calorimetry. In an insulated system, the total heat gained equals the total heat lost: $\sum Q = 0$, counting heat gained as positive and heat lost as negative.

WORKED EXAMPLE 12.2

Quenching hot iron

A 0.20 kg iron bolt at 300°C is dropped into 1.0 kg of water at 20°C in an insulated container. Find the final temperature.

Solution

Heat lost by the iron equals heat gained by the water:

$$0.20(450)(300 - T) = 1.0(4186)(T - 20).$$

$$27\,000 - 90T = 4186T - 83\,720 \Rightarrow 4276T = 110\,720 \Rightarrow T = 25.9^\circ\text{C}.$$

Evaluate. The water's large heat capacity means it barely warms up, while the iron cools by $274\text{ }^{\circ}\text{C}$. That is why water makes such a good coolant, in car engines, in power stations, and in the blood that carries heat from the muscles to the skin.

WORKED EXAMPLE 12.3

From ice to steam

How much energy is needed to turn 1.0 kg of ice at $-10\text{ }^{\circ}\text{C}$ into steam at $100\text{ }^{\circ}\text{C}$?

Solution

Add up each stage:

STAGE	CALCULATION	ENERGY (KJ)
Warm the ice, $-10 \rightarrow 0\text{ }^{\circ}\text{C}$	$1.0(2100)(10)$	21
Melt the ice at $0\text{ }^{\circ}\text{C}$	$1.0(3.34 \times 10^5)$	334
Warm the water, $0 \rightarrow 100\text{ }^{\circ}\text{C}$	$1.0(4186)(100)$	419
Boil the water at $100\text{ }^{\circ}\text{C}$	$1.0(2.26 \times 10^6)$	2260
Total		3030

Evaluate. Boiling takes three-quarters of the total energy, more than five times the energy needed to heat the water from freezing to boiling. This is why steam burns are so severe: steam releases its large latent heat as it condenses on the skin. It is also why sterilising autoclaves use steam, and why evaporating sweat cools the body so effectively.

IN PRACTICE

Thermoregulation. At rest the body produces about 100 W of heat. During hard exercise it can produce more than 1000 W. It sheds this heat by radiation, convection and conduction while the skin is warmer than its surroundings, and by **evaporating sweat**, which is the only route that still works when the air is hotter than the skin. Each kilogram of sweat that evaporates removes about 2.4 MJ. Getting rid of 600 W this way needs $600/2.4 \times 10^6 = 2.5 \times 10^{-4}$ kg/s, almost a litre an hour. In humid air sweat cannot evaporate, and heat stroke becomes a danger. During a fever, the body's thermostat in the hypothalamus is reset higher, and shivering and constriction of blood vessels in the skin drive the temperature up to the new set point.

§12.4

Heat transfer

Conduction

Heat flows through a material from hot to cold, carried by molecular collisions (and, in metals, by free electrons). For a slab of area A , thickness L , and faces at temperatures differing by ΔT , Fourier's law gives

$$P = \frac{dQ}{dt} = kA \frac{\Delta T}{L}.$$

k is the **thermal conductivity**, in W/(m K). Typical values: copper 400, steel 50, glass 0.8, brick 0.7, water 0.6, wood 0.1, fat 0.2, fibreglass insulation 0.04, still air 0.025.

Layers in series. Define each layer's **thermal resistance** $R = L/(kA)$. Then $P = \Delta T/R_{\text{total}}$, with $R_{\text{total}} = R_1 + R_2 + \dots$. This is exactly like electrical resistors in series (Chapter 17). Builders quote the “R-value” L/k , per unit area.

Convection

Heat is carried by the bulk motion of a fluid: warm air rising, or blood flowing. For a surface of area A at temperature T in a fluid at T_a , the heat transfer is often modelled as $P = hA(T - T_a)$, where h is a heat-transfer coefficient. This leads to **Newton's law of cooling**: a body's temperature difference from its surroundings decays exponentially (Problem P12.15).

Radiation

Every body emits electromagnetic radiation. A body at absolute temperature T with surface area A radiates a power given by the **Stefan–Boltzmann law**:

$$P = e\sigma AT^4, \quad \sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}.$$

The **emissivity** e ranges from 0 (perfect reflector) to 1 (perfect absorber, a “black body”). Human skin has $e \approx 0.97$ in the infrared. A body at T in surroundings at T_0 exchanges a net power

$$P_{\text{net}} = e\sigma A(T^4 - T_0^4).$$

The spectrum peaks at a wavelength given by **Wien’s law**:

$$\lambda_{\text{max}}T = 2.898 \times 10^{-3} \text{ m K}.$$

WORKED EXAMPLE 12.4

Why double glazing works

A window has an area of 2.0 m^2 , and the temperature difference across it is 20 K . Compare the conductive heat loss through (a) a single pane of glass 4.0 mm thick and (b) a double-glazed unit made of two such panes separated by a 10 mm gap of still air.

Solution

Work with the thermal resistance per unit area, L/k :

- Glass pane: $0.004/0.8 = 0.005 \text{ m}^2\text{K/W}$.
- Air gap: $0.010/0.025 = 0.40 \text{ m}^2\text{K/W}$.

(a) Single pane:

$$P = \frac{A \Delta T}{L/k} = \frac{2.0(20)}{0.005} = 8000 \text{ W}.$$

(b) Double glazing:

$$P = \frac{2.0(20)}{0.005 + 0.40 + 0.005} = \frac{40}{0.41} = 98 \text{ W}.$$

Evaluate. The thin layer of still air does almost all of the insulating. In reality the single pane loses far less than 8 kW , because thin films of still air cling to both surfaces and add their own resistance (about $0.17 \text{ m}^2\text{K/W}$ in total), giving roughly 230 W . Double glazing still

cuts the loss severalfold. Animals use the same principle: fur, feathers and blubber trap still air or fat.

WORKED EXAMPLE 12.5

Radiating body heat

An unclothed person with a surface area of 1.8 m^2 and a skin temperature of 33°C stands in a room whose walls are at 20°C . Find (a) the net power lost by radiation, taking $e = 0.97$, and (b) the wavelength at which the skin's emission peaks.

Solution

(a) In kelvin, $T = 306 \text{ K}$ and $T_0 = 293 \text{ K}$.

$$P_{\text{net}} = (0.97)(5.67 \times 10^{-8})(1.8)(306^4 - 293^4) = (9.90 \times 10^{-8})(8.768 \times 10^9 - 7.370 \times 10^9)$$

(b) Wien's law:

$$\lambda_{\text{max}} = \frac{2.898 \times 10^{-3}}{306} = 9.5 \times 10^{-6} \text{ m} = 9.5 \mu\text{m} \text{ (far infrared).}$$

Evaluate. The radiative loss alone exceeds the resting metabolic rate (about 100 W), which is why we need clothing or heating even at “room temperature”. Infrared thermometers and thermal cameras detect this $\sim 10 \mu\text{m}$ radiation, and are used for fever screening and to spot inflammation or poor circulation.

§12.5

The ideal gas and kinetic theory

The ideal gas law

At low density, every gas obeys

$$pV = nRT = NkT,$$

where:

- n is the number of moles, and $R = 8.314 \text{ J/(mol K)}$ is the gas constant;
- N is the number of molecules, and $k = R/N_A = 1.381 \times 10^{-23} \text{ J/K}$ is Boltzmann's constant;

- $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ is Avogadro's number.

At standard temperature and pressure (0 °C, 1 atm), one mole of any ideal gas occupies 22.4 L.

Pressure from molecular motion

Consider N molecules, each of mass m , in a cubical box of side L . A molecule with x -velocity v_x bounces elastically off a wall perpendicular to x :

- At each collision its momentum changes by $2mv_x$.
- It returns to the same wall every $2L/v_x$.
- So the average force it exerts on that wall is $\frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}$.

Summing over all the molecules and dividing by the wall's area L^2 :

$$p = \frac{Nm\langle v_x^2 \rangle}{V}.$$

The motion is random, so $\langle v_x^2 \rangle = \frac{1}{3} \langle v^2 \rangle$. Therefore

$$pV = \frac{1}{3} Nm \langle v^2 \rangle = \frac{2}{3} N \langle K_{\text{tr}} \rangle.$$

Comparing this with $pV = NkT$:

KEY IDEA

Temperature measures molecular kinetic energy.

$$\langle K_{\text{tr}} \rangle = \frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} kT, \quad v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}.$$

At a given temperature, all gases have the same average translational kinetic energy per molecule. Lighter molecules simply move faster.

Equipartition of energy. Each independent quadratic term in a molecule's energy (each “degree of freedom”) holds, on average, $\frac{1}{2} kT$. The internal energy of n moles of an ideal gas is therefore:

- **monatomic** gas (3 translational degrees of freedom): $U = \frac{3}{2} nRT$;
- **diatomic** gas near room temperature (3 translational + 2 rotational): $U = \frac{5}{2} nRT$.

The internal energy of an ideal gas depends **only on its temperature**.

WORKED EXAMPLE 12.6

How fast are air molecules?

Find the rms speed of nitrogen molecules ($M = 0.028$ kg/mol) at 300 K, and their average translational kinetic energy.

Solution

$$v_{\text{rms}} = \sqrt{\frac{3(8.314)(300)}{0.028}} = \sqrt{2.67 \times 10^5} = 517 \text{ m/s.}$$

$$\langle K \rangle = \frac{3}{2}kT = \frac{3}{2}(1.381 \times 10^{-23})(300) = 6.2 \times 10^{-21} \text{ J.}$$

Evaluate. Air molecules move faster than the speed of sound (343 m/s). That makes sense: sound is carried *by* molecular motion, so it cannot travel faster than the molecules themselves do.

WORKED EXAMPLE 12.7

How long does a scuba tank last?

A 12 L scuba tank is filled with air to 200 bar at 20 °C. (a) How many moles of air does it hold? (b) What volume would this air occupy at the surface (1.013 bar, 20 °C)? (c) A diver at 30 m depth (4.0 bar absolute) breathes 15 times a minute, 0.50 L per breath at that pressure. Roughly how long will the tank last? Ignore the reserve that must be kept, and assume the air stays at 20 °C.

Solution

(a) Moles:

$$n = \frac{pV}{RT} = \frac{(2.0 \times 10^7)(0.012)}{8.314(293)} = 98.5 \text{ mol.}$$

(b) At the surface, $V = nRT/p = 98.5(8.314)(293)/1.013 \times 10^5 = 2.37 \text{ m}^3$, about 2400 L.

(c) At 4.0 bar, each 0.50 L breath contains the same amount of air as $0.50 \times 4.0/1.013 = 2.0$ L at the surface. The diver therefore uses about 30 surface litres per minute, and

$$t \approx \frac{2370 \text{ L}}{30 \text{ L/min}} \approx 80 \text{ min.}$$

Evaluate. At four times the pressure, each breath contains four times as many molecules, so a tank that would last over 2.5 hours at the surface lasts only a quarter as long at 30 m.

Divers plan their dives around exactly this calculation.

§12.6

The first law of thermodynamics

KEY IDEA

First law. The change in a system's internal energy equals the heat added *to* it minus the work done *by* it:

$$\Delta U = Q - W.$$

This is conservation of energy, extended to include heat.

Work done by a gas. When a gas expands against a piston, it does work

$$W = \int_{V_1}^{V_2} p dV,$$

which is the **area under the curve on a pV diagram**. Work depends on the path taken between two states, and so does heat. Their difference, ΔU , depends only on the end states, because U is a *state function*.

Four important processes for an ideal gas

PROCESS	CONDITION	WORK W	HEAT Q	ΔU
Isochoric	V constant	0	$nC_V\Delta T$	$nC_V\Delta T$
Isobaric	p constant	$p\Delta V$	$nC_p\Delta T$	$nC_V\Delta T$
Isothermal	T constant	$nRT \ln(V_2/V_1)$	$= W$	0
Adiabatic	$Q = 0$	$-\Delta U$	0	$nC_V\Delta T$

Molar heat capacities. At constant volume, all the heat goes into internal energy, so $C_V = \frac{3}{2}R$ for a monatomic gas and $\frac{5}{2}R$ for a diatomic gas. At constant pressure, the gas also does work as it

expands, so it needs more heat:

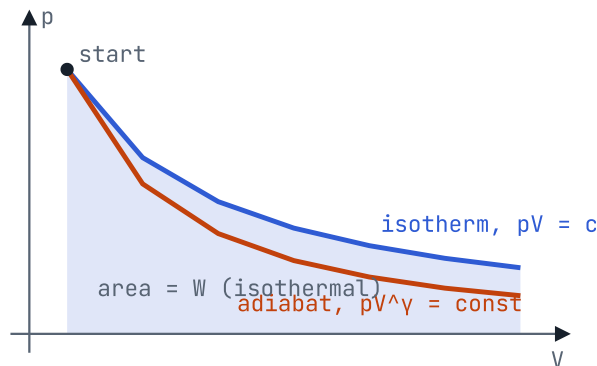
$$C_p = C_V + R \quad (\text{Mayer's relation}).$$

Their ratio is $\gamma = C_p/C_V$, which is $\frac{5}{3}$ for a monatomic gas and $\frac{7}{5} = 1.40$ for a diatomic gas.

Adiabatic processes. These happen too fast for heat to flow (sound waves, engine strokes) or in insulated systems. For an ideal gas:

$$pV^\gamma = \text{constant}, \quad TV^{\gamma-1} = \text{constant}.$$

On a pV diagram, an adiabat is **steeper** than an isotherm through the same point. Compressing a gas adiabatically heats it, because all the work done on it goes into internal energy.



Isothermal expansion (blue) and adiabatic expansion (orange) from the same starting state. The adiabat is steeper because the gas cools as it expands. The shaded area is the work done in the isothermal expansion.

WORKED EXAMPLE 12.8

Isothermal expansion and diesel ignition

(a) Two moles of an ideal gas expand isothermally at 300 K from 10 L to 20 L. Find W , Q and ΔU . (b) A diesel engine compresses air (a diatomic gas, $\gamma = 1.40$) adiabatically from 300 K to one-eighteenth of its original volume. Find the final temperature.

Solution

(a) For an isothermal process $\Delta U = 0$, so

$$W = nRT \ln \frac{V_2}{V_1} = 2(8.314)(300) \ln 2 = 3.46 \times 10^3 \text{ J}, \quad Q = W = 3.46 \text{ kJ}.$$

All the heat absorbed is converted into work.

(b) Using $TV^{\gamma-1} = \text{constant}$:

$$T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1} = 300(18)^{0.40} = 300(3.18) = 953 \text{ K (680 } ^\circ\text{C)}.$$

Evaluate. This temperature is well above the auto-ignition temperature of diesel fuel (about 210 °C), so the fuel ignites the instant it is sprayed in, with no spark plug needed. The high compression ratio is also why diesel engines are more efficient than petrol engines (Problem P12.13).

§12.7

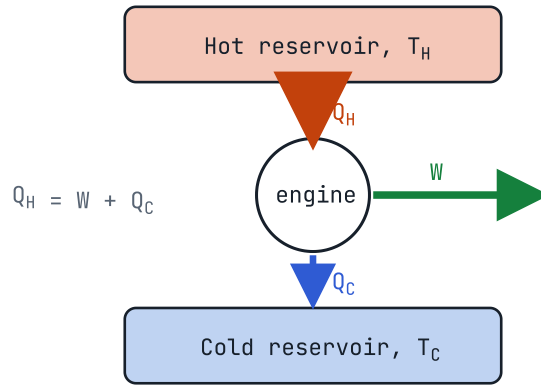
The second law: engines, refrigerators and entropy

The first law says energy is conserved. It does not say which way processes go. Heat flows from hot to cold, never the reverse on its own. A dropped egg does not reassemble. The **second law** captures this one-way character.

KEY IDEA

Second law of thermodynamics, in two equivalent statements:

- **Kelvin–Planck:** no cyclic device can convert heat from a single reservoir entirely into work. Every heat engine must reject some heat to a colder reservoir.
- **Clausius:** heat cannot flow by itself from a colder body to a hotter one. A refrigerator needs work to pump heat “uphill”.



A heat engine takes heat Q_H from a hot reservoir, converts part of it into work W , and must reject the rest, Q_C , to a cold reservoir.

Heat engines. An engine running in a cycle has $\Delta U = 0$ over each cycle, so $W = Q_H - Q_C$. Its efficiency is

$$\eta = \frac{W}{Q_H} = 1 - \frac{Q_C}{Q_H}.$$

The Carnot limit. The most efficient possible engine operating between reservoirs at T_H and T_C is a reversible engine, the *Carnot engine*. For it, $Q_C/Q_H = T_C/T_H$, so

$$\eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H}.$$

No real engine can exceed this. Efficiency improves with a hotter source or a colder sink.

Refrigerators and heat pumps run the cycle in reverse: they use work W to move heat Q_C from cold to hot. Their performance is measured by the **coefficient of performance (COP)**:

$$\text{refrigerator: } K = \frac{Q_C}{W} \leq \frac{T_C}{T_H - T_C}, \quad \text{heat pump: } K_{\text{HP}} = \frac{Q_H}{W} \leq \frac{T_H}{T_H - T_C}.$$

The COP can exceed 1. A heat pump delivers more heat than the electrical energy it uses, because most of that heat is drawn from outdoors.

Entropy. For a reversible transfer of heat dQ at temperature T , the entropy change is $dS = dQ/T$. The second law then says:

KEY IDEA

The total entropy of an isolated system never decreases: $\Delta S_{\text{total}} \geq 0$. It stays constant only for reversible processes. Microscopically, entropy measures disorder: the number of ways a system's molecules can be arranged consistent with its macroscopic state.

WORKED EXAMPLE 12.9

Power stations and heat pumps

(a) A power station's steam enters the turbine at 550 °C, and the condenser is at 30 °C. What is the maximum possible efficiency? Real stations achieve about 40%. Where does the rest of the energy go? (b) A heat pump warms a house to 20 °C while drawing heat from outdoor air at 0 °C. What is its maximum possible COP? Real units achieve 3–4. Compare with an electric heater. (c) Find the entropy change when 1.0 kg of ice melts at 0 °C.

Solution

(a) $T_H = 823$ K and $T_C = 303$ K, so

$$\eta_{\text{Carnot}} = 1 - \frac{303}{823} = 0.63 \text{ (63\%)}.$$

At a real efficiency of 40%, about 60% of the fuel's energy is rejected as heat to the river, the sea or the cooling towers. That is a requirement of the second law, not an engineering failure.

(b) $K_{\text{HP,max}} = \frac{293}{293 - 273} = 14.7$. A real COP of 3.5 means 3.5 kW of heat for every 1 kW of electricity. An electric heater gives exactly 1 kW of heat for 1 kW of electricity (COP = 1). Heat pumps are therefore several times more efficient.

(c) $\Delta S = \frac{Q}{T} = \frac{3.34 \times 10^5}{273} = 1.22 \times 10^3$ J/K. Ordered ice becomes disordered liquid, and the entropy rises.

IN PRACTICE

Thermodynamics at work. Car engines run at about 25–35% efficiency, combined-cycle gas power stations at about 60%, and the best fuel cells, which are not heat engines and so are not bound by the Carnot limit, at about 60% before heat recovery. Medical MRI magnets are superconducting and must be kept at 4 K by liquid helium, maintained by refrigerators fighting an enormous temperature ratio. Living organisms are not heat engines. Muscle converts chemical energy directly into work at about 20–25% efficiency. They do obey the second law: they keep their internal order (low entropy) by exporting entropy to their surroundings as heat and waste.

CHAPTER SUMMARY

- Zeroth law: thermal equilibrium defines temperature. Use kelvin in all thermodynamic formulas.
- Expansion: $\Delta L = \alpha L_0 \Delta T$ and $\Delta V = \beta V_0 \Delta T$, with $\beta \approx 3\alpha$. Thermal stress $\sigma = Y\alpha\Delta T$.
- $Q = mc\Delta T$ and $Q = mL$. Calorimetry: $\sum Q = 0$.
- Conduction: $P = kA\Delta T/L$, with resistances adding in series. Radiation: $P = e\sigma AT^4$, net $e\sigma A(T^4 - T_0^4)$, and $\lambda_{\max}T = 2.898 \times 10^{-3}$ m K.
- Ideal gas: $pV = nRT = NkT$. Kinetic theory: $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}kT$, so $v_{\text{rms}} = \sqrt{3RT/M}$.
- $U = \frac{3}{2}nRT$ (monatomic) or $\frac{5}{2}nRT$ (diatomic). $C_p = C_v + R$, $\gamma = C_p/C_v$.
- First law: $\Delta U = Q - W$, with $W = \int p dV$. Isothermal: $W = nRT \ln(V_2/V_1)$. Adiabatic: pV^γ and $TV^{\gamma-1}$ are constant.
- Engines: $\eta = W/Q_H \leq 1 - T_C/T_H$. Refrigerator COP $\leq T_C/(T_H - T_C)$. Heat pump COP $\leq T_H/(T_H - T_C)$.
- Entropy: $dS = dQ_{\text{rev}}/T$, and $\Delta S_{\text{total}} \geq 0$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P12.1

Why does running hot water over a tight metal lid on a glass jar help to loosen it?

P12.2

Why does steam at 100 °C cause a much more serious burn than the same mass of water at 100 °C?

P12.3

A container holds a mixture of hydrogen and oxygen gas at the same temperature. Compare the average translational kinetic energies of the two kinds of molecule, and their rms speeds.

P12.4

Can you cool a kitchen by leaving the refrigerator door open? Explain using energy conservation and the second law.

LEVEL B – STANDARD PROBLEMS

P12.5

A steel ring with an inner diameter of 4.990 cm at 20 °C must be shrink-fitted onto a shaft of diameter 5.000 cm. To what temperature must the ring be heated to slip over the shaft? ($\alpha_{\text{steel}} = 12 \times 10^{-6} \text{ K}^{-1}$.)

P12.6

50 g of ice at 0 °C is dropped into 300 g of water at 30 °C in an insulated cup. Does all the ice melt? Find the final temperature.

P12.7

Fever. During an infection, a 70 kg patient's temperature rises from 37.0 °C to 39.0 °C. Take the body's average specific heat as 3500 J/(kg K). (a) How much extra internal energy does the body store? (b) If the body generates an extra 50 W of heat above what it loses, how long does the rise take?

P12.8

A 20 m² wall consists of 20 cm of brick ($k = 0.70 \text{ W/(m K)}$) and 10 cm of fibreglass insulation ($k = 0.040$). The inside is at 20 °C and the outside at −5 °C. (a) Find the rate of heat loss. (b) Compare it with the rate for the brick alone. (c) Find the temperature at the boundary between the brick and the insulation, if the insulation is on the inside.

P12.9

The Sun's spectrum peaks at about 502 nm. (a) Estimate the Sun's surface temperature. (b) Find the power radiated per square metre of its surface, assuming it is a black body. (c) The Sun's radius is $6.96 \times 10^8 \text{ m}$. Find its total power output.

P12.10

A car tyre is at a gauge pressure of 200 kPa at 10 °C. After a long drive its temperature is 40 °C. Assuming its volume stays constant, find the new gauge pressure. (Atmospheric pressure is 101.3 kPa.)

P12.11

Find the rms speeds of hydrogen ($M = 0.002 \text{ kg/mol}$) and oxygen ($M = 0.032 \text{ kg/mol}$) molecules at 300 K. Compare them with the Earth's escape speed (11.2 km/s), and explain why the Earth's atmosphere has almost no free hydrogen.

P12.12

A diatomic ideal gas expands at a constant pressure of $1.0 \times 10^5 \text{ Pa}$ from 2.0 L to 5.0 L. Find (a) the work done by the gas, (b) the change in its internal energy, and (c) the heat it absorbs.

LEVEL C – CHALLENGE PROBLEMS**P12.13**

The Otto cycle. An idealised petrol engine cycle consists of an adiabatic compression from volume V_1 to V_2 , heating at constant volume (combustion), an adiabatic expansion from V_2 back to V_1 , and cooling at constant volume (exhaust). Show that its efficiency is $\eta = 1 - r^{1-\gamma}$, where $r = V_1/V_2$ is the compression ratio. Evaluate η for $r = 10$ (petrol) and $r = 18$ (diesel), with $\gamma = 1.4$, and comment on why real engines achieve only about half these values.

P12.14

One mole of air ($\gamma = 1.40$), initially at 300 K and 1.0 atm, is compressed to one-tenth of its volume (a) isothermally and (b) adiabatically. For each case find the final temperature, the final pressure, and the work done on the gas. Explain why the adiabatic compression needs more work.

P12.15

Estimating time of death. Forensic scientists use Newton's law of cooling,

$T - T_a = (T_0 - T_a)e^{-kt}$. A body is found in a room at 20 °C. At 10:00 its core temperature is 30.0 °C, and at 11:00 it is 28.0 °C. Assuming the body was at 37.0 °C at the time of death, estimate when death occurred. What are the main limitations of this method?

P12.16

Heat lost by breathing. In very cold weather (0 °C, dry air), a person breathes 7.5 L of air per minute. The exhaled air is at 37 °C and saturated with water vapour, containing 44 mg of water per litre. The density of air is 1.29 g/L, its specific heat is 1005 J/(kg K), and the latent heat of water at body temperature is 2.4×10^6 J/kg. Estimate (a) the power used to warm the inhaled air and (b) the power used to humidify it. Compare the total with the resting metabolic rate (about 100 W).

P12.17

Sun kinks. A continuous welded steel rail, with cross-sectional area 7500 mm², is fixed in place at 10 °C. On a summer day the rail reaches 45 °C. (a) Find the compressive stress and the total compressive force in the rail. (b) Explain why the rail may buckle sideways, and how railway engineers prevent this.

P12.18

A freezer keeps its interior at -18 °C in a kitchen at 25 °C, and must remove 200 J of heat from the interior each second. (a) What is the minimum electrical power needed? (b) The actual COP is 2.5. Find the actual power drawn and the rate at which heat is rejected into the kitchen. (c) Calculate the rate of entropy change of the interior, of the kitchen, and of the universe, and verify that the second law is satisfied.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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